CF15: Noether's Theorem as Invariant Articulation Cost Under Symmetry

Complete Formalism

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Abstract

This paper closes Noether symmetry and conservation as domain-proper realizations of the Primitive Bifurcation Law of the Void Dynamics Model (VDM). A symmetry is not taken primitively as a standard-physics axiom. It is derived as a direction of transformation along which the invariant pays no additional articulation cost to first order. Because the invariant cannot discharge, the articulation burden borne in such a zero-cost direction cannot be removed; because the direction is zero-cost, the invariant cannot increase that burden there either. The burden is therefore fixed. This is the VDM-first derivation of Noether's theorem.

The derivation inherits the stationary-action closure of CF 14 and the metriplectic split of CF 01/CF 02. It first derives the propagation- resistance limb and field-gradient burden density as VDM objects, then derives the Noether charge and Noether current from those objects, and only then identifies the familiar standard-physics formulas as corollaries. Two forms are closed: the particle-sector theorem and the field-sector theorem. The metriplectic bridge is then completed: under pure J-limb evolution, Noether charges are exactly conserved; under M-limb activity, free Noether charges drift through the same M-bracket that generates entropy, $\dot{Q}_\xi = [Q_\xi, \Sigma]_M$, while $\dot{\Sigma} = [\Sigma, \Sigma]_M \geq 0$.

The major conserved quantities of physics — energy, momentum, angular momentum, electric charge, colour charge, and the stress-energy tensor — are then rebuilt in dependency order as realized descendants of the same zero-cost mechanism. Standard physics appears only as recognition language attached after the VDM derivation is complete.

Keywords Noether's theorem · conservation laws · invariant articulation cost · symmetry · zero-cost direction · metriplectic · primitive bifurcation · void debt · J-limb · M-limb

1 Introduction

1.1 Constitutional Source

CF 15 is the forced next step after CF 14. Once CF 14 closes stationary action as the temporal effective invariant, the next question is immediate: what happens in directions of transformation that cost the invariant nothing at all? If the invariant pays zero articulation cost in some direction, then it cannot increase burden there, because doing so would require a cost that is absent; and it cannot decrease burden there, because that would be discharge. The burden in that direction is therefore fixed.

That is the source of the present paper. Conservation is not introduced as an additional primitive law. It is non-discharge expressed on the temporal path domain along a zero-cost axis. The paper therefore derives what the standard literature calls Noether's theorem rather than assuming it.

1.2 Canon Sources and What They Provide

Every proof in this paper is required to start from canon already earned in the CF stack. Standard physics is never used as a proof step.

A(-1) [1]
Licenses: primitive law of non-discharge; two-pole opposition; same-domain saturation forces orthogonal re-articulation.

CF000 [2]
Licenses: Theorem 4.3.1 that non-bearing is contradictory; forced non-flat articulation; orthogonal re-articulation structure; the two terminal poles.

CF00 [3]
Licenses: carrier domain $\mathcal{M}$; smooth variation structure; complex Hilbert bundle; local $U(1)$ redundancy from complex phase structure; π as the unique least-positive ORS half-turn parameter; quotient-physical projected derivative.

CF01 [4]
Licenses: QGT-induced metriplectic split $J \oplus M$; J skew-symmetric, M symmetric positive semi-definite; degeneracy conditions $J \cdot \delta\Sigma = 0$ and $M \cdot \delta\mathcal{I} = 0$; Poisson/J-bracket on each symplectic leaf; equations of motion $\dot{F} = \{F, H\}_J + [F, \Sigma]_M$.

CF02 [5]
Licenses: entropy non-decrease $\frac{d\Sigma}{dt} = [\Sigma, \Sigma]_M \geq 0$; equality only at equilibrium; positive semi-definite M-bracket structure; near-equilibrium Onsager force–flux form.

CF03 [6]
Licenses: tachyonic potential structure $V''(0) < 0$; scaling theorem for hierarchical tachyonic interfaces.

CF06 [7]
Licenses: Fisher–Ruppeiner metric on M-limb trajectory/state space; statistical distinguishability of nearby states.

CF08 [8]
Licenses: domain-wall construction; tachyonic bulk saturation forces boundary-hosted modes; chiral domain-wall fermion resolution.

CF09 [9]
Licenses: ORS phase redundancy realized as gauge-sector zero-cost structure; Berry-connection emergence from local phase redundancy; the $\pm\ell$ multi-mode degeneracy used in the non-abelian extension.

CF12 [10]
Licenses: Einstein–Hilbert action as gravitational articulation-cost functional; diffeomorphism invariance as zero-cost direction at the geometric level; metric variation produces stress accounting at the gravity layer.

CF13 [11]
Licenses: π transcendence via the ORS orbit closure argument; chirality as oriented non-equivalence of the two half-turn completions; Nielsen–Ninomiya obstruction as an orbit-orientation consequence.

CF14 [12]
Licenses: articulation-cost functional $S_{\mathcal{I}}[\gamma] = \int L_{\mathcal{I}}\, dt$; stationarity $\delta S_{\mathcal{I}} = 0$ for endpoint-fixed paths; the two-pole balance law

$$\frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} = \frac{\partial L_{\mathcal{I}}}{\partial q},$$

with the left limb the propagation-resistance limb and the right limb the configuration-burden limb; the Legendre correspondence from $L_{\mathcal{I}}$ to the energy functional on the selected path.

1.3 Series Placement

The dependency chain is:

- CF 000: primitive law, non-discharge, dependence order
- CF 00: carrier, variation structure, phase redundancy

- CF 01, CF 02: metriplectic $J \oplus M$ split and entropy law
- CF 14: stationary action as temporal effective invariant
- CF 15 (this paper): conservation as burden borne in a zero-cost direction

CF 15 connects backward to CF 14 and forward to CF 09, CF 12, the anomaly closure, and the non-abelian SU(N) extension.

1.4 Scope

This paper does four things:

(i) derives zero-cost directions and their conservation law from non-discharge and CF 14,
(ii) derives the Noether charge and current as VDM objects before attaching the familiar standard formulas,
(iii) closes the metriplectic bridge between exact J-limb conservation and M-limb symmetry breaking, and
(iv) rebuilds the major realized conservation laws in dependency order.

2 Inherited Structures

CF 15 inherits the following structures from the prior canon.

A(-1)/CF000 — non-discharge and bearing.
The invariant cannot discharge into either isolated pole. Any admitted articulation must bear the invariant; non-bearing is contradictory. Same-domain saturation forces orthogonal re-articulation.

CF00 — carrier and variation.
The carrier domain $\mathcal{M}$, smooth local variation, the representative bundle, and the local $U(1)$ redundancy of phase have been derived rather than assumed.

CF01 — QGT and metriplectic split.
The antisymmetric QGT sector induces the reversible J-bracket and the symmetric sector induces the dissipative M-bracket. The degeneracy conditions $J \cdot \delta\Sigma = 0$ and $M \cdot \delta\mathcal{I} = 0$ hold.

CF02 — entropy law.
Entropy is non-decreasing along admissible M-limb evolution: $\dot{\Sigma} = [\Sigma, \Sigma]_M \geq 0$, with equality only on the steady-state set.

CF09 — gauge-sector phase structure.
The ORS phase redundancy of the representative fibre generates gauge-sector zero-cost directions; the $\pm\ell$ structure supplies the multi-mode non-abelian extension.

CF12 — geometric zero-cost structure.
At the geometric level, diffeomorphism invariance is a zero-cost direction of the gravitational articulation-cost functional.

CF14 — path selection and two-pole balance.
For endpoint-fixed admissible paths, $\delta S_{\mathcal{I}} = 0$. The selected path satisfies the local balance law

$$\frac{d}{dt}\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} = \frac{\partial L_{\mathcal{I}}}{\partial q}. \tag{1}$$

The left side is the propagation-resistance limb and the right side is the configuration-burden limb.

3 Zero-Cost Directions: VDM Definition and Mechanism

3.1 VDM Derivation of the Propagation-Resistance Limb

The first object needed for CF 15 is not a standard canonical momentum but a VDM-derived burden carrier inherited from CF 14.

Step 1. CF 14 derives the articulation-cost density $L_{\mathcal{I}}$ on the carrier. The path functional

$$S_{\mathcal{I}}[\gamma] = \int_{t_i}^{t_f} L_{\mathcal{I}}(q, \dot{q}, t)\, dt \tag{2}$$

accumulates total articulation burden along the path.

Step 2. The selected path satisfies the two-pole balance law (1). This already splits path variation into two limbs: a propagation-resistance limb and a configuration-burden limb.

Step 3. The propagation-resistance limb is therefore the rate-sensitive burden stored by the cost density at a point on the path. By the CF 14 balance law, the natural local object carrying that burden is

$$P := \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}}. \tag{3}$$

This is not imported from classical mechanics. It is the VDM object that measures how much articulation burden is stored against changes in traversal rate.

Step 4. The selected-path balance law becomes

$$\frac{dP}{dt} = \frac{\partial L_{\mathcal{I}}}{\partial q}. \tag{4}$$

Thus the time-change of the propagation-resistance limb exactly balances the configuration-burden limb on any selected path.

3.2 Definition of Zero-Cost Direction

Definition 1 (Zero-cost direction)**.** A smooth vector field $\xi(q, \dot{q}, t)$ on the admissible configuration space is a *zero-cost direction* of the articulation-cost density $L_{\mathcal{I}}$ if the first-order variation of $L_{\mathcal{I}}$ along ξ is a total time derivative:

$$\delta_\xi L_{\mathcal{I}} := \frac{\partial L_{\mathcal{I}}}{\partial q} \cdot \xi + \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \dot{\xi} = \frac{d}{dt}\Lambda(q, t) \tag{5}$$

for some admissible boundary function Λ. When $\Lambda = 0$, the direction is strictly zero-cost.

Remark 1 (Boundary admissibility). The total-derivative form is required because endpoint-fixed variations in CF 14 are insensitive to boundary contributions. In compact domains, or in noncompact domains with sufficient fall-off, the boundary contribution vanishes in the observable charge. When that fall-off fails, the boundary contribution must be carried as a genuine observable rather than dropped.

3.3 Why Zero-Cost Directions Produce Conservation

The primitive mechanism is now short.

Step 1: By A(-1) and CF 000, the invariant cannot discharge.

Step 2: By CF 14, continuation along the selected path is admitted articulation and must bear the invariant.

Step 3: If a direction ξ is zero-cost, the invariant pays no additional articulation cost to move in that direction.

Step 4: Therefore the invariant cannot increase burden in direction ξ, because doing so would require paying a cost that is absent.

Step 5: The invariant also cannot remove burden from direction ξ, because that would be discharge.

Step 6: Hence the burden borne in direction ξ is fixed along any selected path.

This fixed burden is the content that the standard literature later names a Noether charge.

3.4 Connection to CF 14

CF 14 gives the path-selection law. CF 15 asks what the selected path conserves when deformed through directions that cost the invariant nothing. The conservation theorem is therefore not an extra variational axiom. It is the unavoidable next articulation of the same CF 14 structure.

4 Formal Derivations

4.1 VDM Derivation of the Noether Charge (Particle Sector)

We now derive, rather than define by analogy, the charge carried by a zero-cost direction.

Step 1 (from CF 14). $L_{\mathcal{I}}$ is the articulation-cost density on the carrier and $S_{\mathcal{I}} = \int L_{\mathcal{I}} dt$ accumulates total articulation burden along a path.

Step 2 (from CF 14). The propagation-resistance limb at a point is $P = \partial L_{\mathcal{I}}/\partial \dot{q}$, and selected paths satisfy $dP/dt = \partial L_{\mathcal{I}}/\partial q$.

Step 3 (from A(-1), CF 000, CF 14). On a selected path, the invariant cannot discharge. For a zero-cost direction ξ, the invariant pays no additional articulation cost when the configuration moves in direction ξ.

Step 4. The burden accumulated in direction ξ by the propagation-resistance limb is the contraction $P \cdot \xi$. If the zero-cost condition carries an admissible boundary contribution Λ, that term must be subtracted to remove the non-load-bearing endpoint piece.

Step 5. Therefore the VDM-derived charge carried by direction ξ is

$$Q_\xi := P \cdot \xi - \Lambda = \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \xi - \Lambda. \tag{6}$$

This formula is not imported from canonical mechanics. It is the earned expression for the articulation burden accumulated by the propagation- resistance limb in the zero-cost direction.

Definition 2 (Noether charge)**.** Given a zero-cost direction ξ with admissible boundary term Λ, the *Noether charge* is the VDM-derived burden (6).

4.2 VDM Derivation of the Field-Gradient Burden Density and Noether Current

The field-sector object must be earned the same way.

Step 1. At field level, the articulation-cost functional is

$$S_{\mathcal{I}}[\phi] = \int \mathcal{L}_{\mathcal{I}}(\phi, \partial_\mu \phi, x)\, d^4x. \tag{7}$$

This is the field-sector realization of the same CF 14 temporal burden functional.

Step 2. The field analogue of the propagation-resistance limb is the burden density stored against field-gradient variation:

$$\Pi^\mu := \frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_\mu \phi)}. \tag{8}$$

This is not imported canonical field momentum. It is the VDM field-gradient burden density inherited from the same path-to-field variation logic.

Step 3. For a zero-cost field variation $\delta\phi$ satisfying

$$\delta \mathcal{L}_{\mathcal{I}} = \partial_\mu K^\mu, \tag{9}$$

the burden carried in that direction is the contraction of the field-gradient burden density with the zero-cost variation, corrected by the admissible boundary flux.

Step 4. Hence the VDM-derived local current is

$$j^\mu := \Pi^\mu\, \delta\phi - K^\mu = \frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_\mu \phi)}\, \delta\phi - K^\mu. \tag{10}$$

Again, this is not a standard import. It is the earned local burden density in the zero-cost field direction.

Definition 3 (Noether current)**.** Given a zero-cost field variation $\delta\phi$ with admissible flux term K^μ, the *Noether current* is the VDM-derived local burden density (10).

4.3 Regularity

Assumption 1 (Regularity)**.** $L_{\mathcal{I}}$ is C^2 in $(q, \dot{q}, t)$, the symmetry generator ξ is smooth, and the field density $\mathcal{L}_{\mathcal{I}}$ is C^2 in $(\phi, \partial_\mu\phi, x)$. These are the minimum smoothness conditions inherited from the carrier and variation structure already earned in CF 00.

5 Noether's Theorem: Particle Mechanics Version

Definition 4 (One-parameter symmetry group)**.** A *one-parameter symmetry group* of $L_{\mathcal{I}}$ is a smooth family ϕ_ϵ with $\phi_0 = \mathrm{id}$ such that, for $q_\epsilon(t) = \phi_\epsilon(q(t))$,

$$\frac{d}{d\epsilon}\bigg|_{\epsilon=0} L_{\mathcal{I}}(q_\epsilon, \dot{q}_\epsilon, t) = \frac{d}{dt}\Lambda(q,t). \tag{11}$$

Its infinitesimal generator is $\xi(q,t) = \frac{d}{d\epsilon}\big|_{\epsilon=0} q_\epsilon$.

Theorem 1 (Noether's theorem — particle sector)**.** *Let $q(t)$ be a selected path satisfying the CF 14 balance law* (1)*. Let ξ generate a one-parameter zero-cost family with boundary contribution Λ. Then the charge* (6) *is conserved:*

$$\frac{d}{dt}Q_\xi = 0. \tag{12}$$

Proof. By CF 14, selected paths satisfy $\frac{d}{dt}\big(\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}}\big) = \frac{\partial L_{\mathcal{I}}}{\partial q}$. By Definition 1, the variation of $L_{\mathcal{I}}$ in the zero-cost direction is $\delta_\xi L_{\mathcal{I}} = \dot{\Lambda}$. By Definition 2,

$$Q_\xi = \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \xi - \Lambda.$$

Differentiating and substituting the selected-path balance gives

$$\begin{aligned}\frac{dQ_\xi}{dt} &= \frac{d}{dt}\Big(\frac{\partial L_{\mathcal{I}}}{\partial \dot{q}}\Big) \cdot \xi + \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \dot{\xi} - \dot{\Lambda} \\ &= \frac{\partial L_{\mathcal{I}}}{\partial q} \cdot \xi + \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \dot{\xi} - \dot{\Lambda} \\ &= \delta_\xi L_{\mathcal{I}} - \dot{\Lambda} = 0.\end{aligned}$$

This is exactly the primitive argument of Section 3: the burden in a zero-cost direction cannot be created and cannot be discharged, so it is fixed. ☐

Corollary 1 (Standard physics recognition)**.** *For a standard Lagrangian system, Theorem 1 recovers the usual mechanical Noether theorem: every continuous one-parameter symmetry of the action produces a conserved quantity.*

6 Noether's Theorem: Field-Theoretic Version

Remark 2 (Field-level balance law from the CF 14 stationarity principle). CF 14 derives the particle-sector two-pole balance law from stationarity of $S_{\mathcal{I}}[\gamma]$. Applying the same stationarity principle to the field-sector functional

$$S_{\mathcal{I}}[\phi] = \int \mathcal{L}_{\mathcal{I}}\, d^4x$$

with boundary-fixed field variations $\delta\phi = 0$ on $\partial\Omega$ gives

$$\delta S_{\mathcal{I}} = \int \left[\frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_\mu \phi)}\right)\right] \delta\phi\, d^4x = 0.$$

Since $\delta\phi$ is arbitrary in the bulk, the bracket must vanish pointwise. Therefore the field-level two-pole balance law is

$$\partial_\mu\Big(\frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_\mu \phi)}\Big) - \frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial \phi} = 0. \tag{13}$$

This is the field-sector analog of the CF 14 stationarity closure, derived by direct application of the same principle to the field functional.

Theorem 2 (Noether's theorem — field sector)**.** *Let ϕ satisfy the field Euler–Lagrange balance law derived from the stationarity of $S_{\mathcal{I}}[\phi]$. Let $\delta\phi$ be a zero-cost field variation with admissible boundary flux K^μ so that $\delta\mathcal{L}_{\mathcal{I}} = \partial_\mu K^\mu$. Then the current* (10) *is locally conserved:*

$$\partial_\mu j^\mu = 0. \tag{14}$$

Proof. By Remark 2, the field balance law is (13). From Definition 3,

$$j^\mu = \frac{\partial \mathcal{L}_\mathcal{I}}{\partial(\partial_\mu \phi)}\,\delta\phi - K^\mu.$$

Taking the divergence and using the product rule gives

$$\partial_\mu j^\mu = \partial_\mu\Big(\frac{\partial \mathcal{L}_\mathcal{I}}{\partial(\partial_\mu \phi)}\Big)\delta\phi + \frac{\partial \mathcal{L}_\mathcal{I}}{\partial(\partial_\mu \phi)}\partial_\mu(\delta\phi) - \partial_\mu K^\mu.$$

But the zero-cost condition is exactly

$$\delta\mathcal{L}_\mathcal{I} = \frac{\partial \mathcal{L}_\mathcal{I}}{\partial \phi}\,\delta\phi + \frac{\partial \mathcal{L}_\mathcal{I}}{\partial(\partial_\mu \phi)}\partial_\mu(\delta\phi) = \partial_\mu K^\mu.$$

Substituting this into the divergence formula yields

$$\partial_\mu j^\mu = \Big[\partial_\mu\Big(\frac{\partial \mathcal{L}_\mathcal{I}}{\partial(\partial_\mu \phi)}\Big) - \frac{\partial \mathcal{L}_\mathcal{I}}{\partial \phi}\Big]\delta\phi = 0$$

on shell by (13). Thus the local burden carried in the zero-cost field direction has no source and no sink. ☐

Corollary 2 (Standard physics recognition)**.** *For an ordinary classical field theory, Theorem 2 recovers the usual local current-conservation form of Noether's theorem.*

Remark 3 (Boundary-supported charges). When the spatial domain is noncompact and the fall-off of j^μ is insufficient, the conserved observable is the total burden carried by the bulk plus the boundary flux. The boundary term is then physical rather than a nuisance term.

7 The Metriplectic Bridge

Theorem 3 (J-limb exact conservation)**.** *On a symplectic leaf where the system evolves under pure J-limb flow, any Noether charge Q_ξ of the Hamiltonian layer is exactly conserved:*

$$\frac{d}{dt}Q_\xi = \{Q_\xi, H\}_J = 0. \tag{15}$$

Proof. Step 1 (from CF 01): the metriplectic equation of motion for any admissible observable F is

$$\frac{dF}{dt} = \{F, H\}_J + [F, \Sigma]_M.$$

Step 2 (Hamiltonian-layer zero-cost condition): by Definition 4, a symmetry direction is one whose first-order variation is zero-cost up to an admissible boundary term, and by Definition 1 that means the invariant pays no first-order burden cost along that direction. Applying this to the Hamiltonian layer, the first-order variation of the Hamiltonian burden vanishes along the symmetry flow generated by Q_ξ. In the J-bracket representation, that is exactly

$$\{Q_\xi, H\}_J = 0.$$

This is the Hamiltonian-layer analogue of the same zero-cost condition: the invariant pays no first-order burden cost in direction Q_ξ at that layer.

Step 3: under pure J-limb evolution, M-limb activity is absent and Σ is frozen, so the second term vanishes:

$$[Q_\xi, \Sigma]_M = 0.$$

Substituting $F = Q_\xi$ into the CF 01 equation of motion gives

$$\frac{dQ_\xi}{dt} = \{Q_\xi, H\}_J + [Q_\xi, \Sigma]_M = 0 + 0 = 0.$$

The J-limb is the reversible sector inherited from CF 01: it cannot create or destroy burden, and with the M-limb absent no second mechanism remains to move Q_ξ. Therefore Q_ξ is exactly conserved on the leaf. ☐

Theorem 4 (M-limb symmetry breaking — general law)**.** *For a Hamiltonian Noether charge Q_ξ in a metriplectic system,*

$$\frac{d}{dt}Q_\xi = \{Q_\xi, H\}_J + [Q_\xi, \Sigma]_M. \tag{16}$$

If Q_ξ is a symmetry of H, then

$$\frac{d}{dt}Q_\xi = [Q_\xi, \Sigma]_M. \tag{17}$$

At the same time,

$$\frac{d\Sigma}{dt} = [\Sigma, \Sigma]_M \geq 0. \tag{18}$$

Thus free Noether-charge drift and entropy production are generated by the same irreversible operator.

Proof. The equation of motion for any observable is inherited from CF 01: $\dot{F} = \{F, H\}_J + [F, \Sigma]_M$. Applying this to $F = Q_\xi$ gives the first formula. If Q_ξ is a Hamiltonian symmetry, then Theorem 3 gives $\{Q_\xi, H\}_J = 0$, leaving only the M-bracket term. The entropy law (18) is exactly the CF 02 second-law closure. Hence the same M-operator that advances entropy is the operator that breaks free Noether conservation away from equilibrium. □

Corollary 3 (Aligned M-limb breaking)**.** *If on the relevant irreversible sector*

$$[Q_\xi, \Sigma]_M = \kappa_\xi(x)\,[\Sigma, \Sigma]_M, \tag{19}$$

then

$$\frac{d}{dt}Q_\xi = \kappa_\xi(x)\,\frac{d\Sigma}{dt}. \tag{20}$$

In the normalized aligned case $\kappa_\xi = 1$, the charge drifts exactly at the entropy-production rate.

Proof. Substitute (19) into (17) and use (18). □

Definition 5 (Irreversible vector field and common Noether kernel)**.** Write the M-bracket as $[F, G]_M = M^{\mu\nu}\partial_\mu F\,\partial_\nu G$. The irreversible entropy-producing vector field is

$$v_M^\mu := M^{\mu\nu}\partial_\nu \Sigma. \tag{21}$$

For the set of free Noether charges $\mathcal{Q}_N^{\text{free}}$, define

$$\mathcal{K}_x := \bigcap_{Q_\xi \in \mathcal{Q}_N^{\text{free}}} \ker(dQ_\xi|_x) \subset T_x\mathcal{M}. \tag{22}$$

Theorem 5 (Equilibrium symmetry maximisation)**.** *Suppose the system reaches a steady state with $[\Sigma, \Sigma]_M = 0$. Then the irreversible sector is inert, and every Hamiltonian Noether charge is exactly conserved. Equilibrium therefore realizes the maximal symmetry compatible with the Hamiltonian layer.*

Proof. By CF 02, $[\Sigma, \Sigma]_M = 0$ is exactly the equilibrium condition. By Definition 5,

$$v_M^\mu = M^{\mu\nu}\partial_\nu \Sigma.$$

In coordinates,

$$[\Sigma, \Sigma]_M = M^{\mu\nu}\partial_\mu \Sigma\,\partial_\nu \Sigma.$$

By CF 01, $M^{\mu\nu}$ is symmetric positive semi-definite. For a positive semi-definite quadratic form, $x^T M x = 0$ implies $Mx = 0$. Therefore $[\Sigma, \Sigma]_M = 0$ forces

$$M^{\mu\nu}\partial_\nu \Sigma = 0,$$

which is exactly

$$v_M^\mu = 0$$

at equilibrium.

Again by Definition 5, $[F, \Sigma]_M = dF(v_M)$ for every admissible observable F. Since $v_M = 0$ at equilibrium, it follows that

$$[F, \Sigma]_M = 0$$

for every admissible F, and in particular for every free Noether charge Q_ξ.

Theorem 4 then gives

$$\frac{dQ_\xi}{dt} = [Q_\xi, \Sigma]_M = 0,$$

while Theorem 3 already gives the vanishing of the J-bracket term for Hamiltonian Noether charges. Hence every free Noether charge of H is exactly conserved at equilibrium. Equilibrium therefore restores exact conservation of the full Hamiltonian Noether sector. □

Lemma 1 (Transversality equivalence)**.** *At any x in the active non-equilibrium set, $[Q_\xi, \Sigma]_M(x) = 0$ for all free Noether charges if and only if $v_M(x) \in \mathcal{K}_x$.*

Proof. By Definition 5, $[Q_\xi, \Sigma]_M = dQ_\xi(v_M)$. The simultaneous vanishing of all such pairings is therefore exactly the condition that v_M lies in the intersection of the corresponding kernels. □

Definition 6 (Closed Noether sector)**.** A realized sector is *closed* with respect to the free Noether charges if all non-Casimir zero-cost directions admitted at that structural level have been realized and no additional hidden articulation axes remain in the sector beyond them.

Lemma 2 (Sector completeness)**.** *In a closed realized sector, every admissible non-Casimir invariant-bearing zero-cost channel at that structural level is represented in the free Noether set $\mathcal{Q}_N^{\text{free}}$.*

Proof. Every admissible zero-cost direction generates a conserved burden by Theorems 1 and 2. Charges that are M-Casimirs are excluded from the free set by structural immunity, not omission. If an admissible non-Casimir zero-cost direction were absent, the sector would be incompletely bearing the invariant and would violate the closure condition. □

Theorem 6 (Boundary-hosted non-equilibrium symmetry reduction)**.** *Let the sector be closed, and let B be an open set on which the M-limb is actively carrying unresolved burden, i.e. $[\Sigma, \Sigma]_M(x) > 0$ for all $x \in B$. Then at least one free Noether charge has nonzero M-drift at some point of B.*

Proof. Assume the contrary: that every free Noether charge has zero M-drift on all of B. By Lemma 1, the irreversible vector field then lies in the common Noether kernel throughout B. But by hypothesis the M-limb is nontrivially carrying unresolved burden there. If that burden does not pass through any admitted free zero-cost channel, then either the invariant is advancing without being borne in any admitted channel, contradicting non-bearing prohibition from CF 000, or the flow is using a hidden channel, contradicting Lemma 2. Thus some free Noether charge must have nonzero M-drift on B. □

8 Realized Conservation Laws in Dependency Order

Each subsection now begins with the VDM structural mechanism and only then states the standard-physics recognition.

8.1 Energy — Zero-Cost Temporal Translation

At the CF 14 path layer, time translation is a zero-cost direction when the articulation-cost density has no explicit time burden. The propagation- resistance limb contracted with the temporal generator therefore yields the conserved burden of the temporal direction. By the CF 14 Legendre correspondence, that burden is the path energy functional:

$$Q_{\partial_t} = \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \dot{q} - L_{\mathcal{I}}. \tag{23}$$

Corollary 4 (Standard physics recognition)**.** *This recovers conservation of the Hamiltonian/energy for a time-translation invariant system.*

8.2 Momentum — Zero-Cost Carrier Translation

Remark 4 (Carrier translation uniformity from invariant indifference)*.* CF 00 derives the carrier $\mathcal{M}$ as the generic host forced by the invariant's own structure. The invariant bears unresolved two-pole opposition and contains no location-distinguishing content: it does not privilege one point of the carrier over another. Translational uniformity at the particle-path layer is therefore not an extra geometric axiom but a direct consequence of the invariant's indifference to position. Absent additional burden-breaking structure, no carrier point receives higher or lower first-order articulation cost than any other.

Movement of the configuration by a constant carrier displacement therefore does not change first-order burden. The carrier displacement direction $\hat{n}$ is zero-cost, so the burden borne by the propagation-resistance limb in that direction is conserved:

$$Q_{\hat{n}} = \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot \hat{n}. \tag{24}$$

Corollary 5 (Standard physics recognition)**.** *In the translationally uniform carrier realization, this is linear momentum conservation in the direction $\hat{n}$.*

8.3 Angular Momentum — Zero-Cost Carrier Rotation

Remark 5 (Carrier rotational uniformity from invariant indifference)*.* CF 00 derives the carrier $\mathcal{M}$ as the generic host of an invariant that contains no orientation-distinguishing content. The unresolved two-pole opposition has no preferred axis, handed direction, or absolute orientation at the carrier level. Rotational uniformity of the articulation-cost structure is therefore a direct consequence of the invariant's indifference to orientation: absent additional burden-breaking structure, rigid reorientation of the configuration does not alter first-order articulation cost.

Rigid rotation about an axis $\hat{n}$ is therefore a zero-cost direction. The burden borne by the propagation-resistance limb in that rotational direction is conserved:

$$Q_{\hat{n}} = \frac{\partial L_{\mathcal{I}}}{\partial \dot{q}} \cdot (\hat{n} \times q). \tag{25}$$

Corollary 6 (Standard physics recognition)**.** *In the rotationally isotropic carrier realization, this recovers angular momentum conservation about axis $\hat{n}$.*

8.4 Electric Charge — ORS Phase Redundancy

CF 09 closes the key mechanism: the invariant cannot distinguish absolute phase along the ORS rotation redundancy of the representative fibre. That phase direction is therefore zero-cost at the gauge layer. Applying the field-sector theorem to the infinitesimal phase variation $\delta\phi = i\phi$ gives the conserved local burden current

$$j^{\mu}_{\mathrm{EM}} = \frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_{\mu}\phi)}(i\phi) + \mathrm{h.c.} \tag{26}$$

with local conservation on shell.

Corollary 7 (Standard physics recognition)**.** *This recovers the usual conserved $U(1)$ electric current and charge.*

Remark 6 (π and charge quantization)*.* The ORS orbit closes at 2π, with π the unique least positive half-turn parameter earned from the ORS structure in CF 00 and refined in CF 13. Charge quantization is therefore read as the winding-number closure of a closed phase orbit rather than as an extra postulate.

8.5 Colour Charge — Non-Abelian Fibre Rotation

The non-abelian proof begins from the CF 09 multi-mode degeneracy rather than from the explicit downstream connection formula. When the fibre carries N degenerate modes, the invariant cannot distinguish rotations between those degenerate orientations that preserve the fibre norm and quotient-physical content. Those traceless fibre rotations are therefore zero-cost directions.

Proposition 1 (SU(N) generators are zero-cost directions)**.** *For the realized N-mode fibre sector, each generator direction T^a ($a = 1, \ldots, N^2 - 1$) is a zero-cost direction of the invariant.*

Proof. By CF 09, the fibre admits multi-mode degeneracy. Rotating between degenerate fibre orientations preserves the quotient-physical content already earned from CF 00. Hence the articulation-cost density is unchanged to first order along each traceless fibre-rotation generator. By Definition 1, each such direction is zero-cost. □

Applying Theorem 2 to the infinitesimal variation $\delta\phi = iT^a\phi$ yields the conserved local burdens

$$j^{\mu,a} = \frac{\partial \mathcal{L}_{\mathcal{I}}}{\partial(\partial_{\mu}\phi)}(iT^a\phi) + \mathrm{h.c.} \tag{27}$$

with covariant conservation once the connection sector is included.

Corollary 8 (Standard physics recognition)**.** *This recovers the non-abelian colour currents and their covariant conservation law. The explicit coordinate connection is a downstream corollary of the CF09 SU(N) extension, not a proof input here.*

8.6 Stress-Energy — Diffeomorphism Zero-Cost Structure

CF 12 closes the gravitational mechanism: at the geometric layer, the Einstein–Hilbert articulation-cost functional is insensitive to coordinate relabeling of the same physical configuration. Diffeomorphism variation is therefore a zero-cost direction at the gravity layer. The corresponding local burden current is the metric-variation stress accounting of the gravitational and matter sector,

$$T^{\mu\nu} := \frac{-2}{\sqrt{-g}} \frac{\delta(\sqrt{-g}\,\mathcal{L})}{\delta g_{\mu\nu}}, \tag{28}$$

with covariant conservation as the geometric realization of the same zero-cost mechanism.

Corollary 9 (Standard physics recognition)**.** *This recovers the stress-energy tensor and the covariant conservation law* $\nabla_\mu T^{\mu\nu} = 0$.

Table 1: Realized descendants of the zero-cost mechanism.

Realized layer	VDM zero-cost mechanism	Standard-physics recognition
Temporal path layer	no first-order time-burden change	energy
Carrier translation	no first-order burden change under uniform displacement	momentum
Carrier rotation	no first-order burden change under isotropic rotation	angular momentum
Gauge phase sector	ORS phase redundancy invisible to the invariant	electric charge
Multi-mode fibre sector	degenerate traceless fibre rotations invisible	colour charge
Geometric layer	diffeomorphism relabeling is zero-cost	stress-energy conservation

9 Inverse Noether and the Symmetry–Conservation Duality

Proposition 2 (Inverse Noether direction)**.** *On a nondegenerate symplectic leaf under pure J-limb evolution, every conserved observable* Q *with* $\{H, Q\}_J = 0$ *generates a one-parameter symmetry flow of the Hamiltonian layer and, via the CF 14 Legendre correspondence, of the selected-path action.*

Proof. In the pure J-limb sector, reversible evolution is generated entirely by the J-bracket. If $\{H, Q\}_J = 0$, then motion generated by Q leaves the Hamiltonian burden unchanged. That means the J-flow generated by Q is a zero-cost direction of the Hamiltonian layer. By the CF 14 correspondence between path burden and Hamiltonian burden on the selected path, the same flow acts as a symmetry of the action functional. □

Remark 7 (Scope of the duality). The symmetry–conservation bijection holds on nondegenerate symplectic leaves of the pure J-sector. Degenerate Casimir directions and M-limb activity lie outside that strict bijection for structural reasons already encoded in CF 01/CF 02.

10 Forward Predictions and Program Map

P1. Every admitted zero-cost direction produces a conserved burden.

P2. Pure J-limb evolution exactly preserves Noether charges.

P3. M-limb activity breaks free Noether conservation through the same operator that produces entropy.

P4. Equilibrium maximizes the conserved Hamiltonian symmetry content, whereas active non-equilibrium reduces it on its support.

P5. The full SU(N) connection formulas of the non-abelian sector are coordinate realizations of the already-closed zero-cost fibre theorem.

P6. The stress-energy conservation law is not primitive but the geometric Noether realization of diffeomorphism zero-cost structure.

P7. Broken symmetries in downstream sectors correspond to an obstruction of a zero-cost direction by irreversible flow, boundary concentration, or the failure of closure conditions at that structural level.

11 Falsification Criteria

CF 15 is falsified if any of the following occur:

F1. A zero-cost direction satisfying (5) fails to produce a conserved burden on a selected path.

F2. A quantity is exactly conserved on a pure J-limb nondegenerate leaf but generates no J-symmetry direction.

F3. A Noether charge drifts under pure J-limb evolution while its zero-cost condition remains satisfied.

F4. A steady state with $[\Sigma, \Sigma]_M = 0$ still exhibits free M-induced Noether drift.

F5. A field configuration satisfying the field balance law yields $\partial_\mu j^\mu \neq 0$ for a genuine zero-cost field direction.

F6. A free Noether charge drifts under M-limb activity without corresponding nonnegative entropy production.

12 Validation Gates

Gate: G1: Zero-cost mechanism *Threshold: Closed.*

Derive conservation from non-discharge plus a zero-cost direction.
Status: Closed in Section 3 and Theorem 1.

Gate: G2: Particle-sector theorem *Threshold: Closed.*

Derive the conserved charge from the selected-path balance law and the zero-cost condition.
Status: Closed in Theorem 1.

Gate: G3: Field-sector theorem *Threshold: Closed.*

Derive local current conservation from the field balance law and the zero-cost field variation.
Status: Closed in Theorem 2.

Gate: G4: J-limb exact conservation *Threshold: Closed.*

Show exact conservation of Noether charges under pure reversible flow.
Status: Closed in Theorem 3.

Gate: G5: M-limb symmetry breaking *Threshold: Closed.*

Show that free Noether charges drift through the entropy-generating M-operator.
Status: Closed in Theorem 4.

Gate: G6: Equilibrium maximization and active non-equilibrium reduction *Threshold: Closed.*

Show that equilibrium restores all Hamiltonian Noether charges, and that an active closed non-equilibrium sector must break at least one free Noether charge.
Status: Closed in Theorem 5, Lemma 1, Lemma 2, and Theorem 6.

Gate: G7: Classical realized charges *Threshold: Closed.*

Recover energy, momentum, and angular momentum from the zero-cost mechanism at the realized carrier/path layers.
Status: Closed in Section 8.

Gate: G8: U(1) charge conservation *Threshold: Closed.*

Recover electric charge from ORS phase redundancy.
Status: Closed in Section 8.

Gate: G9: Stress-energy conservation *Threshold: Closed.*

Recover stress accounting and covariant conservation from diffeomorphism zero-cost structure.
Status: Closed in Section 8.

Gate: G10: SU(N) colour charge conservation *Threshold: Closed.*

Recover non-abelian conserved currents from multi-mode fibre degeneracy.
Status: Closed via Proposition 1 and Section 8.

Gate: G11: Anomaly as chiral orbit non-closure and domain-wall resolution *Threshold: Closed.*

Carry the anomaly as a boundary consequence of oriented chiral orbit non-closure and boundary hosting.
Status: Closed in Discussion by inheritance from CF13 and CF08.

Gate: G12: Tachyonic vacuum splitting and the Higgs normal mode *Threshold: Closed.*

Derive broken-vacuum tangential zero-cost modes and the normal restoring mode at the saturated interface.
Status: Closed in Discussion by inheritance from CF03, CF08, CF09.

Gate: G13: Fluctuation relations from entropy geometry *Threshold: Closed.*

Derive fluctuation identities from the CF02 entropy law and the CF06 Fisher–Ruppeiner geometry.
Status: Closed in Discussion by inheritance from CF02 and CF06.

13 Discussion

13.1 Why the Derivation Is Forced

The entire paper reduces to one point: the invariant cannot create burden in a direction it does not pay for, and cannot remove burden from a direction without discharge. A zero-cost direction therefore carries fixed burden. All standard conservation formulas are later recognitions of that one mechanism.

13.2 The Variational Spine Is Now Complete

CF 14 closes the path-selection law. CF 15 closes what the selected path conserves and how that conservation is broken by the irreversible limb. Together they form the complete variational spine of the present stage of the formalism.

13.3 Anomaly as Boundary-Hosted Chiral Non-Closure

The VDM structure comes first. CF 13 closes chirality as the oriented non-equivalence of the two half-turn completions of the ORS orbit. A compact carrier cannot globally host both chiral completions without an obstruction. CF 08 then provides the forced resolution: when bulk hosting saturates, the unresolved chiral burden is carried on a domain wall. The anomaly is therefore a boundary-hosted failure of global chiral orbit closure rather than an inexplicable correction term.

Corollary 10 (Standard physics recognition)**.** *This recovers the chiral anomaly and its domain-wall fermion resolution.*

13.4 Tachyonic Vacuum Saturation and the Higgs Normal Mode

The VDM derivation again starts structurally. By CF 03, a tachyonic origin with $V''(0) < 0$ cannot stably host the invariant in the symmetric bulk. Saturation forces re-articulation to a new vacuum interface, and CF 08 turns that interface into a domain wall. Tangential directions along the new vacuum orbit remain zero-cost and therefore carry conserved burdens; the normal direction away from the interface costs burden and is therefore restoring. The normal mode is the massive interface mode of the saturated vacuum.

Corollary 11 (Standard physics recognition)**.** *This recovers spontaneous symmetry breaking, Goldstone modes, and the Higgs scalar as the normal mode of the vacuum interface.*

13.5 Fluctuation Relations from M-Limb Entropy Geometry

The VDM-first statement is that the M-limb assigns asymmetric weight to a trajectory and its reverse according to the entropy borne along that trajectory. CF 02 provides the nonnegative entropy-production law, and CF 06 provides the Fisher–Ruppeiner distinguishability geometry on the relevant trajectory/state space. Those two structures already force exact identities between forward and reverse ensemble weights.

Corollary 12 (Standard physics recognition)**.** *This recovers the Crooks fluctuation theorem and the Jarzynski equality as ensemble identities of the M-limb entropy geometry.*

13.6 Boundary Charge as a Physical Observable

When bulk hosting fails to close a current because the domain is noncompact or the fall-off is insufficient, the unresolved burden cannot disappear. By non-discharge, it must remain borne. The correct observable is then the total of the bulk burden and the boundary-hosted flux burden. Boundary charge is therefore physical rather than a disposable remainder.

14 Conclusions

CF 15 derives Noether's theorem from the Primitive Bifurcation Law. A symmetry is a zero-cost direction of the articulation-cost structure. Because the invariant cannot discharge and cannot pay articulation cost where none exists, the burden borne in a zero-cost direction is fixed. That is the entire conservation mechanism.

The paper first derives the relevant burden carriers — the propagation- resistance limb and the field-gradient burden density — as VDM objects, and only then derives the charge and current formulas built from them. The particle-sector theorem and the field-sector theorem are both closed without using standard physics as a proof step. The metriplectic bridge is then closed: J-limb evolution exactly conserves Hamiltonian Noether charges, while M-limb activity breaks free Noether conservation through the same operator that generates entropy. Equilibrium restores the full Hamiltonian symmetry content, while active non-equilibrium in a closed sector must break at least one free Noether charge.

The major realized conservation laws — energy, momentum, angular momentum, electric charge, colour charge, and stress-energy conservation — are all rebuilt in dependency order as descendants of the same zero-cost mechanism. Together with CF 14, this paper closes the variational spine of the current VDM formalism: CF 14 derives the selection principle, and CF 15 derives what it conserves and what breaks it.

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